

Question 1 **1 mark**

101

Manitoba postal codes consist of three letters and three digits. Determine the total number of possible postal codes in Manitoba if the following characteristics must be met:

- The postal code must begin with the letter R.
- Only 18 letters of the alphabet can be used.
- The letters and digits must alternate.
- The letters and digits can be repeated.

Question 2 **2 marks**

102

The percent of people who click on a link, in response to seeing an advertisement on YouTube, can be modeled by the formula:

$$R(t) = -(0.8)^{0.2t} + 0.6$$

where $R(t)$ is the percentage, in decimal form, of people who click on the link

t is the time in days.

Determine, algebraically, the number of days it will take for 45% of people who see the advertisement to click on the link.

Question 3 **3 marks**

103

Determine and simplify the middle term in the binomial expansion of $\left(\frac{5}{x} + 4x^3\right)^6$.

Question 4 **3 marks**

104

Solve $3\tan^2x + 5\tan x - 6 = 0$, over the interval $[0, 2\pi]$.

Note: A calculator is not required for the remaining test questions.

Question 5

a) 1 mark b) 1 mark

105
106

Given $\theta = -80^\circ$,

a) determine the measure of θ , in radians.

b) state all the coterminal angles of θ .

Question 6

1 mark

107

State a possible value of n if the polynomial function $p(x) = (x+1)^n (x-4)^2$ has a range of $(-\infty, \infty)$.

Question 7

4 marks

108

Describe the transformations used to obtain the graph of the function $y = -2f(x + 8) - 5$ from the graph of $y = f(x)$.

Question 8

2 marks

109

State the domain and range of the function $f(x) = \frac{-2}{x^2}$.

Domain: _____

Range: _____

Question 9

1 mark

110

Given the following row of Pascal's Triangle, state the values in the next row.

1 5 10 10 5 1

Question 10

2 marks

111

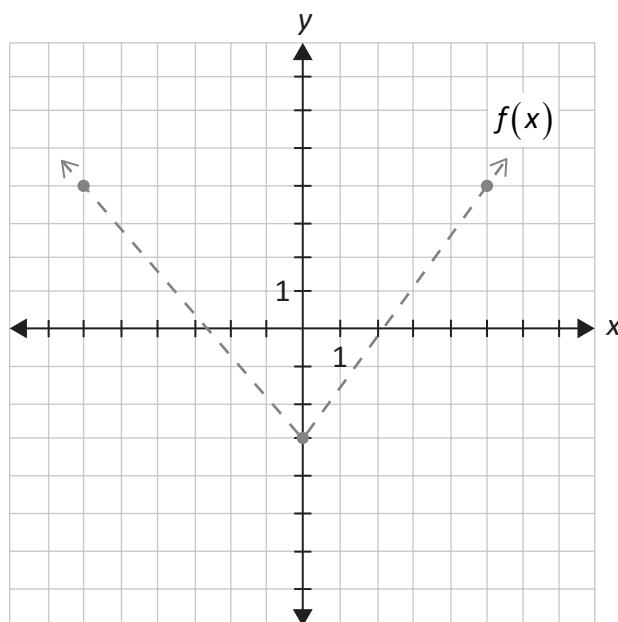
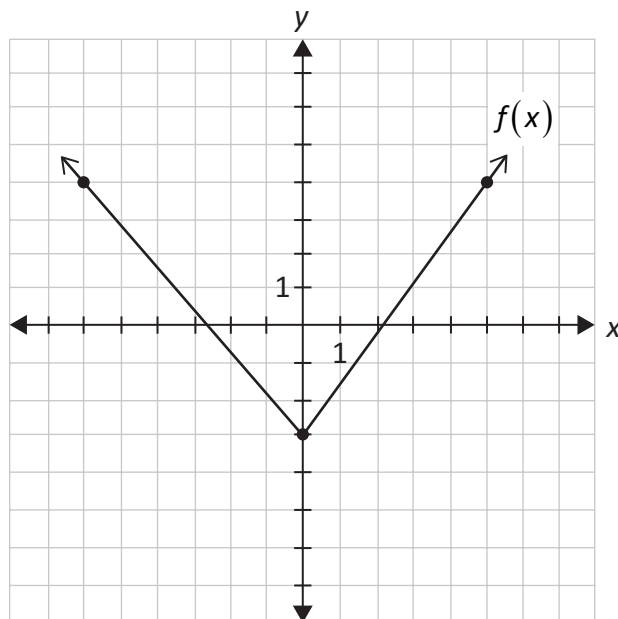
Given $f(x) = 3 - 2x$, determine the equation of $f^{-1}(x)$.

Question 11

2 marks

112

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 12

3 marks

113

Prove the identity for all permissible values of x .

$$\frac{\csc^2 x + \sec^2 x}{\tan x + \cot x} = \csc x \sec x$$

Left-Hand Side	Right-Hand Side

Question 13

1 mark

114

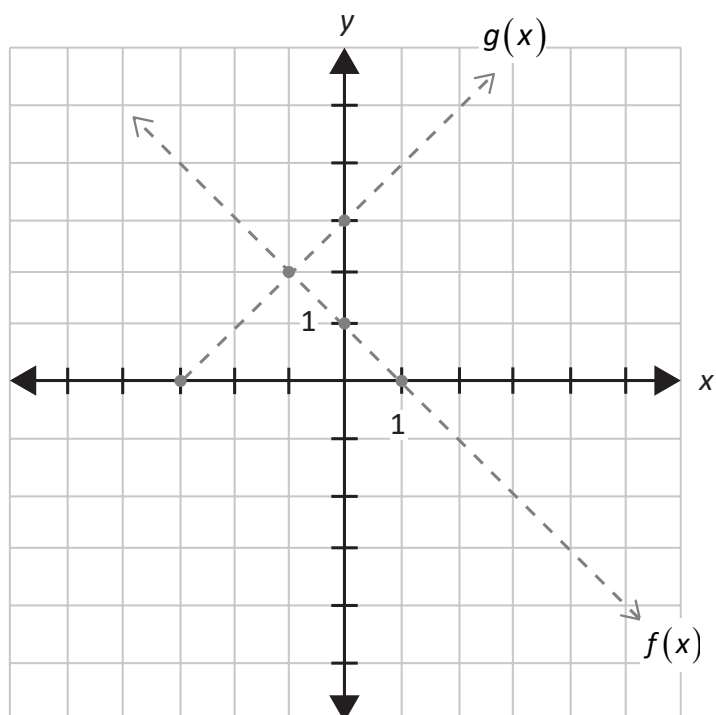
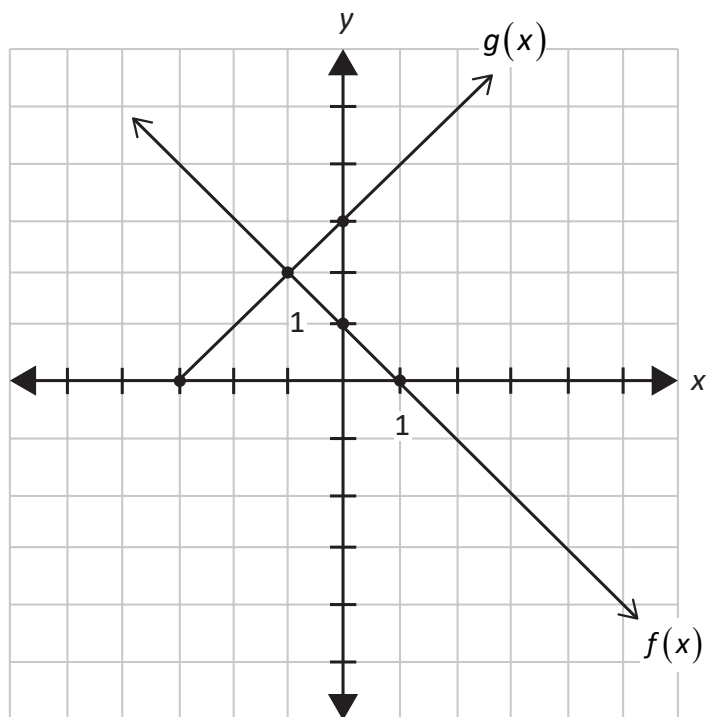
Justify that the graph of $f(x) = e^{x+1} + 5$ does not have an x -intercept.

Question 14

2 marks

115

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) \cdot g(x)$.



The graph of $f(x)$ and $g(x)$ have already been drawn for your reference.

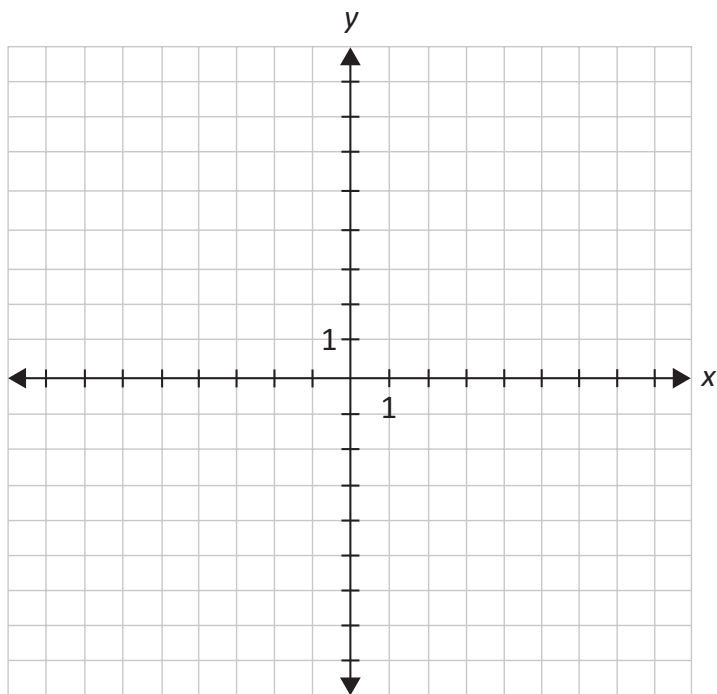
No marks will be awarded for the graph of $f(x)$ and $g(x)$.

Question 15

4 marks

116

Sketch the graph of $f(x) = \frac{5}{2}\sqrt{-(x-4)}$.



NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.

